Acta Tropica 190 (2019) 312–319

Contents lists available at ScienceDirect

Acta Tropica

journal homepage: www.elsevier.com/locate/actatropica

Wing morphometric analysis of forensically important flesh flies (Diptera: Sarcophagidae) in Thailand

Narin Sontigun[a], Chutharat Samerjai[a], Kom Sukontason[a], Anchalee Wannasan[a], Jens Amendt[b], Jeffery K. Tomberlin[c], Kabkaew L. Sukontason[a,*]

[a] Department of Parasitology, Faculty of Medicine, Chiang Mai University, Chiang Mai, 50200, Thailand
[b] Institute of Legal Medicine, Forensic Biology/Entomology, Kennedyallee 104, 60596 Frankfurt am Main, Germany
[c] Department of Entomology, Texas A&M University, College Station, TX, USA

ARTICLE INFO

Keywords:
Flesh fly
Wing
Morphometry
Forensic
Entomology
Thailand

ABSTRACT

Flesh flies are insects of forensic importance as the larvae associated with human remains can be used to estimate a minimum post-mortem interval (PMI_{min}) in most cases. And, because life-history traits can vary across species, correct identification is a mandatory step before being used as evidence. Adult flesh flies are extremely similar in general appearance, which causes difficulty in species identification as it is largely based on the morphology of the male genitalia; this also makes it difficult to identify females. Currently, landmark-based geometric morphometric analysis of insect wings has proven to be a valuable tool for species identification. Herein, we applied wing morphometric analysis of 524 flesh fly specimens comprising 12 species from Thailand. The right wing of each specimen was removed, mounted on a microscope slide, photographed, and digitized using 18 landmarks. Wing shape variation among genera and species were analyzed using canonical variate analysis, while wing shape variation between sexes of each species was analyzed using discriminant function analysis. A cross-validation test was used to evaluate the reliability of classification. Results of this study demonstrate wing shape can be used to separate genera and species, and distinguish between sexes of the same species, with high reliability. Therefore, the landmark-based geometric morphometric analysis of wings is a useful additional method for species and sex discrimination of flesh flies.

1. Introduction

Flesh flies are regarded as insects of medical and forensic importance since adults are mechanical carriers of human pathogens (Graczyk et al., 2005; Getachew et al., 2007), while larvae can cause myiasis [e.g., *Parasarcophaga dux* Thompson, *Liopygia ruficornis* (Fabricius), and *Liopygia argyrostoma* Robineau-Desvoidy (Ferraz et al., 2010; Chaiwong et al., 2014; Giangaspero et al., 2017)]. A more significant aspect of medical importance is that some species are recognized to be forensically important as their immature stages are often associated with decomposing human corpses, which can be used to estimate the minimum post-mortem interval (PMI_{min}) in forensic investigations [e.g., *P. dux*, *L. ruficornis*, *Boettcherisca peregrina* (Robineau-Desvoidy), and *L. argyrostoma* (Chigusa et al., 2005; Sukontason et al., 2007; Kavitha et al., 2013)] assuming colonization after death.

In Thailand, 29 flesh fly genera comprised of 83 species within the subfamilies Miltogramminae and Sarcophaginae have been listed (Kurahashi and Chaiwong, 2013). As several species of the subfamily Sarcophaginae have been documented as being synanthropic [e.g., *B. peregrina*, *Boettcherisca nathani* (Lopes), *L. ruficornis*, *P. dux*, *Parasarcophaga brevicornis* Ho, *Parasarcophaga scopariiformis* Senior-White, *Lioproctia pattoni* Senior-White, and *Seniorwhitea princeps* Wiedemann (Bänziger and Pape, 2004; Chaiwong et al., 2012)], and are of great medical and forensic importance. Generally, the feeding habits of larval Sarcophaginae include necrophagy, coproghagy, parasitism, kleptoparasitism, and predation (Pape, 1996; Richet et al., 2011). According to Bänziger and Pape (2004), Sarcophaginae in Thailand are categorized as having three general behaviors, based on their larval habitat preferences: (1) coprobiodotic (larviposition on feces), for example, *Bercaea africa* (Wiedemann), *Parasarcophaga albicep* (Meigen), and *Parasarcophaga misera* (Walker); (2) necrobiodotic (larviposition on carrion), for example, *B. nathani*, *B. peregrina*, and *Lioproctia saprianovae* Pape and Bänziger; and (3) amphibiodotic (larviposition on both feces and carrion), for example, *P. dux*, *P. brevicornis*, *L. ruficornis*, *L. pattoni*, *Sarcorohdendorfia antilope* Böttcher, and *Sarcorohdendorfia multivillosa* (Shinonaga and Tumrasvin). Additionally, larvae of *P. dux* and *L.*

* Corresponding author at: Department of Parasitology, Faculty of Medicine, Chiang Mai University, Chiang Mai, 50200, Thailand.
E-mail address: kabkaew.s@cmu.ac.th (K.L. Sukontason).

https://doi.org/10.1016/j.actatropica.2018.12.011
Received 18 October 2018; Received in revised form 3 December 2018; Accepted 6 December 2018
Available online 07 December 2018
0001-706X/ © 2018 Elsevier B.V. All rights reserved.

ruficornis have been documented as myiasis producers in humans within Thailand (Chaiwong et al., 2014; Chiewchanvit et al., 2017). Among Thai flesh flies, *P. dux*, *L. ruficornis*, and *B. peregrina* are of the greatest forensic importance as they are commonly found associating with human and other vertebrate animal remains (Sukontason et al., 2007, 2010; Vitta et al., 2007; Sukchit et al., 2015).

Since flesh flies are of forensic importance, correct species identification is an important first step in calculating the PMI_{min}. Generally, flesh flies can largely be identified based on the morphology of the external male genitalia (Kurahashi and Chaiwong, 2013; Meiklejohn et al., 2013; Vairo et al., 2014) and molecular characteristics (Tan et al., 2010; Guo et al., 2012; Jordaens et al., 2013). However, because of the high complexity of the structure of the male genitalia, identification is quite difficult for non-taxonomists; also, since identification keys for females are rare (Richet et al., 2011; Vairo et al., 2015), their identification is problematic. Moreover, DNA identification is expensive and requires sophisticated equipment.

In addition to traditional morphological identification, or DNA identification, the landmark-based geometric morphometric analysis of insect wings has proven to be a valuable tool for interspecific discrimination (Jaramillo-O et al., 2015; Sontigun et al., 2017), sexual dimorphism (Gidaszewski et al., 2009; Virginio et al., 2015), and geographic variation of species (Hall et al., 2014; Carvajal et al., 2016). This recognition is because it is simple, fast, inexpensive, and highly reliable. Several dipteran taxa have been studied, such as fruit flies (Drosophilidae) (Bubliy et al., 2008; Perre et al., 2014), mosquitoes (Culicidae) (Jaramillo-O et al., 2015; Wilke et al., 2016), black flies (Simuliidae) (Pepinelli et al., 2013), midges (Ceratopogonidae) (Muñoz-Muñoz et al., 2014), blow flies (Calliphoridae) (Lyra et al., 2010; Sontigun et al., 2017), muscids (Muscidae) (Grzywacz et al., 2017), and tabanids (Tabanidae) (Cárdenas et al., 2013). However, only one report on wing morphometric analysis of flesh fly species has been published to date (Brown and Shipp, 1978). Hence, the aim of this study is to test landmark-based geometric morphometric analysis of wings for species identification of medically and forensically important flesh flies of Thailand.

2. Materials and methods

2.1. Specimen collection

A total of 524 flesh flies were used in this study. They were collected from field and laboratory colonies. For field collections, adults were collected from several localities of Chiang Mai and Chiang Rai provinces of northern Thailand during 2007–2016 (Table 1 and Fig. 1) using a handheld fly net, sweeping it over a bait of 1-day-old beef offal (~250 g) and then killed by ethyl acetate in the field. The specimens were pinned and identified under a stereo microscope (Olympus, Japan) based on the taxonomic key of Kurahashi and Chaiwong (2013), including *B. nathani*, *B. peregrina*, *L. pattoni*, *P. brevicornis*, *P. scoparuiformis*, *S. antilope*, *S. multivillosa*, *Sarcorohdendorfia seniorwhitei* Walker, and *S. princeps* (Table 1). Since the taxonomic key developed by Kurahashi and Chaiwong (2013) relies heavily on the male genitalia, female identification is not possible. Thus, offspring of field-captured gravid females must be reared to the adult stage and resulting males used to confirm the species. Regarding specimens of *L. ruficornis*, *L. saprianovae* and *P. dux*, they were obtained from the laboratory colonies maintained at the Department of Parasitology, Faculty of Medicine, Chiang Mai University, Thailand. Thus, 12 species included in this study covering most of the taxa that are of medical and forensic importance.

2.2. Slide preparation

The right wing of each fly was removed with fine forceps, and then the basicostae were cut off from the wing using a scalpel blade. The mounting procedure was performed as previously described (Sontigun et al., 2017).

2.3. Image processing and data acquisition

Each wing was photographed with an Olympus DP22 digital camera coupled to an Olympus SZX7 stereomicroscope at 1.25× magnification. Images were kept in the same folder, and then the tps file was built using TpsUtil software version 1.64 (http://life.bio.sunysb.edu/morph/) to minimize the possible bias when digitizing landmark locations. Eighteen landmarks as used by Hall et al. (2014) (Fig. 2) were digitized using TpsDig2 software version 2.20 (http://life.bio.sunysb.edu/morph/) by the same person (NS). Each wing was digitized twice by the same person (NS) to reduce measurement error (Arnqvist and Mårtensson, 1998).

2.4. Geometric morphometric analysis

The tps file with raw coordinates of landmarks from two replicates of digitization for each specimen was loaded into MorphoJ software version 1.06 (Klingenberg, 2011), and then superimposed using Procrustes Fit function to eliminate variation in scale, position, and orientation from the raw coordinates. The Procrustes coordinates and the centroid size (the square root of the sum of the squared distances between the center of configuration of the landmarks and each landmark) (Bookstein, 1991) obtained from the landmark data after the Procrustes superimposition were averaged for each specimen to remove any possible measurement error prior to statistical analyses. A significance level of 0.05 was used for determining the significant difference between groups in all subsequent analyses.

2.4.1. Allometry

Because wing size affects wing shape variation (allometry) (Gidaszewski et al., 2009; Sontigun et al., 2017), an estimation of the allometric effects is an important first step before performing the shape analyses. To evaluate the effect of allometry, the regression of the Procrustes coordinates (dependent variable) on the centroid size (independent variable) was analyzed among species, pooled within species, and for each species separately, using a permutation test with 10,000 rounds in MorphoJ software version 1.06 (Klingenberg, 2011). The effect of allometry on sexual shape dimorphism was also analyzed by examining the regression of the Procrustes coordinates, pooled within sex, on the centroid size. When scaling relationships was statistically significant among species or between sexes ($P < 0.05$), residuals from the regression of shape on size were used to test the differences in shape without the size effect (allometry-free variables or size-corrected shapes).

2.4.2. Shape variation

A total of 524 specimens (312 males and 212 females) were used in shape analysis. Wing shape differences between groups (genera or species) were analyzed by canonical variate analysis (CVA) based on Mahalanobis distances and tested with a permutation test (10,000 rounds) using MorphoJ software version 1.06 (Klingenberg, 2011). A cross-validation test in discriminant function analysis (DFA) was used to evaluate the reliability of classification based on the Mahalanobis distances in a permutation test with 10,000 rounds using MorphoJ software version 1.06 (Klingenberg, 2011). As females are much more abundant than males on carcasses (Martín-Vega and Baz, 2013; Szpila et al., 2015), and their identification is problematic due to the limitation of identification keys for females, female specimens of six species (*B. nathani*, *B. peregrina*, *L. pattoni*, *L. saprianovae*, *L. ruficornis*, and *P. dux*) were analyzed separately by CVA and cross-validation test to confirm the reliability of wing shape as regards species identification of females.

Table 1
List of Thai flesh fly specimens used in this study.

Species	Species code	Collection site	District	No. of specimens		
				Males	Females	Total
Boettcherisca nathani	BN	Doi Nang Kaew	Doi Saket*	8	–	8
		Huay Mae Khanin	Hang Dong*	2	–	2
		Tardmok waterfall	Mae Rim*	2	–	2
		Tham Phra Leusri, Doi Suthep-Pui mountain	Mueang Chiang Mai*	11	–	11
		Doi Inthanon	Chom Thong*	–	7	7
		San Ku, Doi Suthep-Pui mountain	Mueang Chiang Mai*	–	10	10
		Sirindhorn Observatory, Doi Suthep-Pui mountain	Mueang Chiang Mai*	–	5	5
	Total			23	22	45
Boettcherisca peregrina	BP	Ban Pong	Hang Dong*	4	–	4
		Chang Puek	Mueang Chiang Mai*	4	–	4
		Doi Kham temple	Mueang Chiang Mai*	3	–	3
		Doi Nang Kaew	Doi Saket*	2	–	2
		Nong Kwai	Hang Dong*	6	–	6
		Ping river	San Sai*	2	–	2
		Huai Mahinfon fall	Mae Chanˆ	–	11	11
		Pa Tueng	Mae Chanˆ	–	5	5
	Total			21	16	37
Lioproctia pattoni	LP	Ban Pang Daeng	Doi Saket*	32	22	54
		San Ku, Doi Suthep-Pui mountain	Mueang Chiang Mai*	–	9	9
	Total			32	31	63
Liopygia ruficornis	LR	Laboratory colony	-*	49	62	111
	Total			49	62	111
Lioproctia saprianovae	LS	Laboratory colony	-*	38	34	72
	Total			38	34	72
Parasarcophaga brevicornis	PB	Doi Kham Temple	Mueang Chiang Mai*	2	–	2
		Elephant camp	Mae Taeng*	2	–	2
		Nang Kaeo	Hang Dong*	3	–	3
		Nong Kwai	Hang Dong*	2	–	2
		Ping river	San Sai*	4	–	4
		Saluang	Mae Rim*	3	–	3
	Total			16	–	16
Parasarcophaga dux	PD	Laboratory colony	-*	46	47	93
	Total			46	47	93
Parasarcophaga scopariiformis	PS	Ban Pong	Hang Dong*	2	–	2
		Huay Tueng Tao	Mae Rim*	7	–	7
		Nong Kwai	Hang Dong*	2	–	2
		Saluang	Mae Rim*	2	–	2
	Total			13	–	13
Sarcorohdendorfia antilope	SA	Doi Nang Kaew	Doi Saket*	15	–	15
		Sirindhorn Observatory, Doi Suthep-Pui mountain	Mueang Chiang Mai*	5	–	5
	Total			20	–	20
Sarcorohdendorfia multivillosa	SM	Doi Indhanon (Chiang Mai)	Chom Thong*	1	–	1
		Doi Nang Kaew	Doi Saket*	10	–	10
		Doi Suthep-Pui, Muang	Mueang Chiang Mai*	2	–	2
	Total			13	–	13
Sarcorohdendorfia seniorwhitei	SS	Doi Nang Kaew	Doi Saket*	3	–	3
		Sirindhorn Observatory, Doi Suthep-Pui mountain	Mueang Chiang Mai*	7	–	7
		Tardmok waterfall	Mae Rim*	1	–	1
		Tham Phra Leusri, Doi Suthep-Pui mountain	Mueang Chiang Mai*	1	–	1
	Total			12	–	12
Seniorwhitea princeps	SP	Ban Pang Daeng	Doi Saket*	6	–	6
		Ban Pong	Hang Dong*	4	–	4
		Doi Kham Temple	Mueang Chiang Mai*	4	–	4
		Tambol Suthep Moo 4	Mueang Chiang Mai*	3	–	3
		Nong Ho	Mueang Chiang Mai*	2	–	2
		Nong Kwai	Hang Dong*	5	–	5
		Pong Yang	Mae Rim*	5	–	5
	Total			29	–	29

*Chiang Mai province.
ˆChiang Rai province.

2.4.3. Sexual shape dimorphism (SShD)

Wing shape differences between males and females of six species (*B. nathani*, *B. peregrina*, *L. pattoni*, *L. saprianovae*, *L. ruficornis*, and *P. dux*) were assessed by DFA based on the Mahalanobis distances and tested in a permutation test with 10,000 rounds using MorphoJ software version 1.06 (Klingenberg, 2011). In this process, a cross-validation test based on the Mahalanobis distances was also performed to evaluate the reliability of the classification between sexes for each species.

3. Results

3.1. Allometry

Regression of Procrustes coordinates on centroid size among species had highly significant differences (permutation test with 10,000 rounds in MorphoJ: $P < 0.0001$); allometry accounted for 1.58% of the total shape variation. Based on the female wings among six species, it can be stated that allometry explained 1.98% of the total shape variation. The

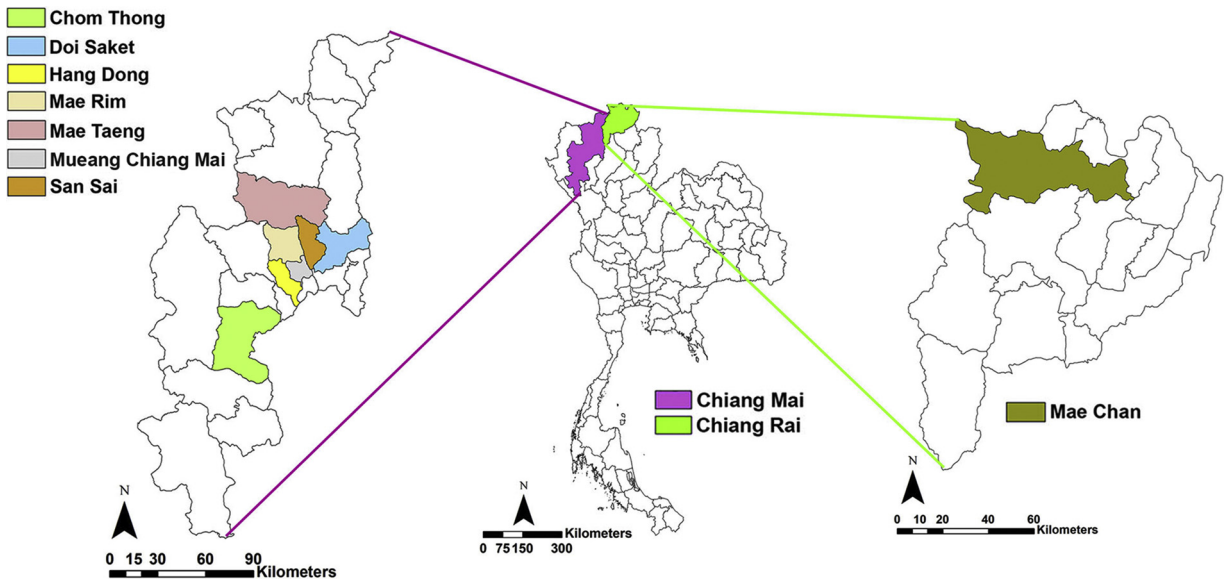

Fig. 1. Map of Thailand showing the collection sites of flesh flies in seven districts (Chom Thong, Doi Saket, Hang Dong, Mae Rim, Mae Taeng, Mueang Chiang Mai, and San Sai) of Chiang Mai province and Mae Chan district of Chiang Rai province, northern Thailand.

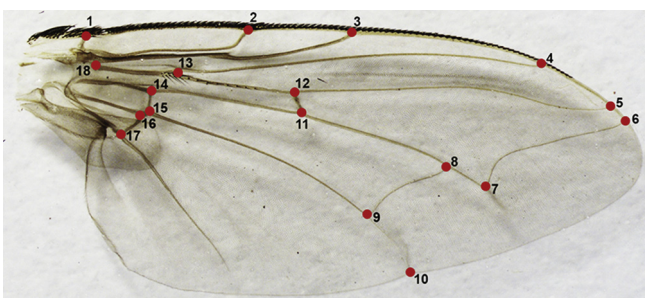

Fig. 2. The right wing of female *B. peregrina* showing the 18 plotted landmarks, based on Hall et al. (2014).

Table 2
Percentage of prediction indicating amount of size-related shape variation of wings in each flesh fly species and between sexes of each species.

Species	% predicted within species	*P*-value	% predicted between sexes	*P*-value
B. nathani	4.4	0.0849	5.9	0.0105
B. peregrina	4.0	0.1693	7.3	0.0056
L. pattoni	15.4	< 0.0001	4.8	0.0058
L. saprianovae	1.4	0.3534	1.7	0.2461
L. ruficornis	3.2	0.0137	5.2	< 0.0001
P. dux	5.1	0.0006	2.0	0.0606
P. brevicornis	20.5	0.0021	–	–
P. scopariiformis	7.6	0.5257	–	–
S. antilope	4.6	0.5298	–	–
S. multivillosa	17.2	0.0381	–	–
S. seniorwhitei	6.5	0.7270	–	–
S. princeps	12.0	0.0012	–	–

Regression of the Procrustes coordinates on the centroid size of wings within each species and between sexes of each species performed using a permutation test with 10,000 rounds in MorphoJ.

relationship between shape and size within each species revealed that allometry was found in six species including *L. pattoni*, *L. ruficornis*, *P. brevicornis*, *P. dux*, *S. multivillosa*, and *S. princeps* (permutation test with 10,000 rounds in MorphoJ: $P < 0.0001$, $P < 0.01$, and $P < 0.05$) (Table 2). The regression of shape as related to size for each species is shown in Supplementary Figure S1. Additionally, the wing shape variation was found to be significantly correlated to the size between sexes in most species (permutation test with 10,000 rounds in MorphoJ: $P < 0.0001$, $P < 0.01$, and $P < 0.05$), except for *L. saprianovae* and *P. dux* (permutation test with 10,000 rounds in MorphoJ: $P > 0.05$) (Table 2).

3.2. Shape variation

After the removal of the allometry, the CVA showed differences in the wing shape among genera and species. At the genus level, the CVA revealed five canonical variates, of which the first two canonical variates (Fig. 3) explained 85.8% of the total variation (CV1 = 51.5%, CV2 = 34.3%). The scatter plot from CV1 and CV2 (Fig. 3) showed that specimens of *Liopygia* were clearly separated into distinct groups, while specimens of the other five genera showed large overlapping over each other. The Mahalanobis distances obtained from the pairwise comparisons of all six genera revealed highly significant differences (permutation test with 10,000 rounds in MorphoJ: $P < 0.0001$), ranging from 4.399 (*Parasarcophaga* and *Seniorwhitea*) to 10.2558 (*Liopygia* and *Seniorwhitea*) (Supplementary Table S1). The percentages of correctly classified specimens obtained from the cross-validation test ranged from 89.7% to 100.0% (Table 3).

At the species level, the CVA revealed 11 canonical variates, of which the first two canonical variates (Fig. 4) explained 73.6% of the total variation (CV1 = 44.7%, CV2 = 28.9%). The scatter plot from CV1 and CV2 (Fig. 4) showed large overlapping among species, except for *L. ruficornis*, which was clearly separated from the other 11 species. The Mahalanobis distances obtained from the pairwise comparisons of the 12 species revealed highly significant differences (permutation test with 10,000 rounds in MorphoJ: $P < 0.0001$), ranging from 2.1316 (*B. nathani* and *B. peregrina*) to 12.1332 (*L. ruficornis* and *P. scopariiformis*) (Supplementary Table S2). The cross-validation test yielded a high percentage of correctly classified specimens in the majority of the comparisons (81.3–100.0%), except for seven pairwise comparisons (*B. nathani* and *B. peregrina*, *P. brevicornis* and *P. scopariiformis*, *P. brevicornis* and *S. princeps*, *P. scopariiformis* and *S. antilope*, *P. scopariiformis* and *S. princeps*, *S. antilope* and *S. multivillosa*, and *S. antilope* and *S. seniorwhitei*) which had percentages of correct classification less than 76% (Table 4).

Based on the female wings of the six species, it can be stated that the CVA revealed five canonical variates, of which the first two canonical variates (Fig. 5) explained 84.6% of the total variation (CV1 = 49.9%,

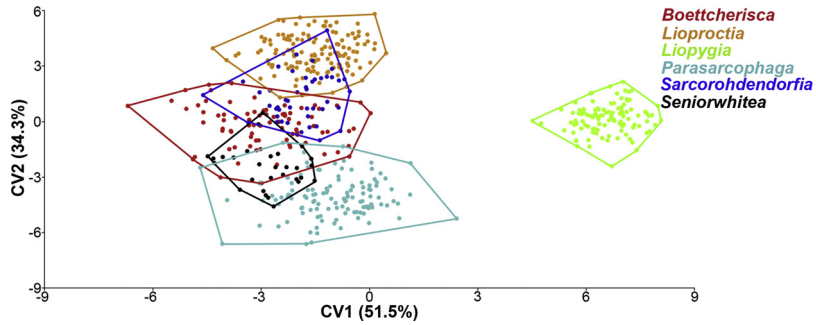

Fig. 3. Scatter plot showing the variation in the shape of wings of six flesh fly genera along the first two canonical variate (CV1 and CV2) axes.

Table 3
Percentage of correct classification obtained from pairwise comparison of analyzed genus with cross-validation test using permutation test with 10,000 rounds in MorphoJ.

		Group 2					
		Boettcherisca	*Lioproctia*	*Liopygia*	*Parasarcophaga*	*Sarcorohdendorfia*	*Seniorwhitea*
Group 1	*Boettcherisca*	–	**100.0**	**100.0**	**98.8**	**97.6**	**96.3**
	Lioproctia	100.0	–	**100.0**	**100.0**	**99.3**	**100.0**
	Liopygia	100.0	100.0	–	**100.0**	**100.0**	**100.0**
	Parasarcophaga	98.4	100.0	100.0	–	**98.4**	**91.0**
	Sarcorohdendorfia	100.0	91.1	100.0	100.0	–	**97.8**
	Seniorwhitea	89.7	100.0	100.0	89.7	93.1	–

The values in bold are the percentages of the correct classification of group 1 obtained from group 1 compared with group 2; the values in narrow are the percentages of the correct classification of group 2 obtained from group 2 compared with group 1.

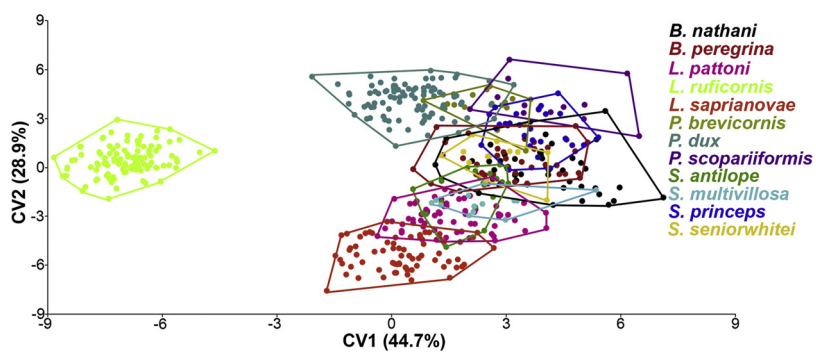

Fig. 4. Scatter plot showing the variation in the shape of wings of 12 flesh fly species along the first two canonical variate (CV1 and CV2) axes.

Table 4
Percentages of correct classification obtained from pairwise comparison of analyzed species with cross-validation test using permutation test with 10,000 rounds in MorphoJ.

		Group 2											
		BN	BP	LP	LR	LS	PB	PD	PS	SA	SM	SP	SS
Group 1	BN	–	**68.9**	**100.0**	**100.0**	**100.0**	**91.1**	**100.0**	**91.1**	**97.8**	**100.0**	**93.3**	**100.0**
	BP	64.9	–	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**94.6**	**89.2**	**97.3**	**94.6**	**100.0**
	LP	100.0	100.0	–	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**95.2**	**100.0**	**100.0**
	LR	100.0	100.0	100.0	–	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**
	LS	100.0	100.0	100.0	100.0	–	**100.0**	**100.0**	**100.0**	**98.6**	**100.0**	**100.0**	**100.0**
	PB	81.3	100.0	100.0	100.0	100.0	–	**93.8**	**68.8**	**93.8**	**100.0**	**62.5**	**87.5**
	PD	98.9	100.0	100.0	100.0	100.0	95.7	–	**98.9**	**98.9**	**100.0**	**100.0**	**100.0**
	PS	84.6	92.3	100.0	100.0	100.0	69.2	92.3	–	**53.9**	**100.0**	**69.2**	**84.6**
	SA	90.0	95.0	100.0	100.0	100.0	95.0	100.0	50.0	–	**75.0**	**100.0**	**70.0**
	SM	92.3	100.0	84.6	100.0	100.0	92.3	100.0	100.0	61.5	–	**100.0**	**92.3**
	SP	82.8	96.6	100.0	100.0	100.0	75.9	96.6	75.9	96.6	89.7	–	**96.6**
	SS	100.0	100.0	100.0	100.0	91.7	83.3	100.0	100.0	66.7	83.3	100.0	–

The values in bold are the percentages of the correct classification of group 1 obtained from group 1 compared with group 2; the values in narrow are the percentages of the correct classification of group 2 obtained from group 2 compared with group 1. Abbreviations: BN, *B. nathani*; BP, *B. peregrina*; LP, *L. pattoni*; LR, *L. ruficornis*; LS, *L. saprianovae*; PB, *P. brevicornis*; PD, *P. dux*; PS, *P. scopariiformis*; SA, *S. antilope*; SM, *S. multivillosa*; SP, *S. princeps*; and SS, *S. seniorwhitei.*

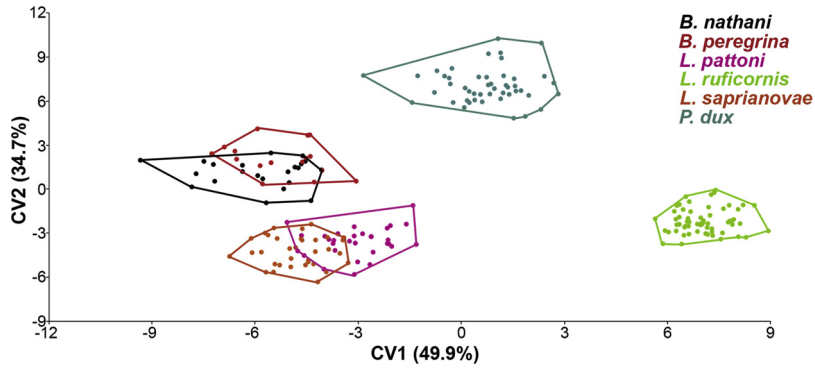

Fig. 5. The scatter plot showing the variation in the shape, based on female wings of six flesh fly species along the first two canonical variate (CV1 and CV2) axes.

Table 5
Percentages of correct classification based on female wings obtained from pairwise comparison of analyzed species with cross-validation test using permutation test with 10,000 rounds in MorphoJ.

		Group 2					
		B. nathani	*B. peregrina*	*L. pattoni*	*L. ruficornis*	*L. saprianovae*	*P. dux*
Group 1	*B. nathani*	–	**77.3**	**100.0**	**100.0**	**100.0**	**100.0**
	B. peregrina	81.3	–	**100.0**	**100.0**	**87.5**	**100.0**
	L. pattoni	100.0	100.0	–	**100.0**	**100.0**	**100.0**
	L. ruficornis	100.0	100.0	100.0	–	**100.0**	**100.0**
	L. saprianovae	100.0	97.1	100.0	100.0	–	**100.0**
	P. dux	100.0	93.6	100.0	100.0	100.0	–

The values in bold are the percentages of the correct classification of group 1 obtained from group 1 compared with group 2; the values in narrow are the percentages of the correct classification of group 2 obtained from group 2 compared with group 1.

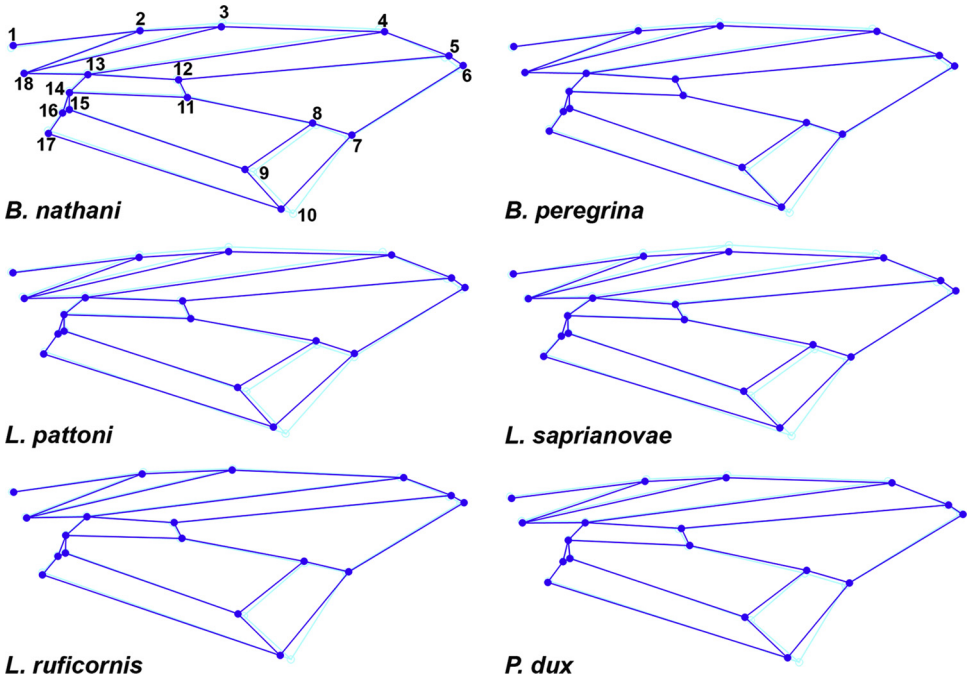

Fig. 6. The wireframe graph showing the differences in the wing shape between males and females of six flesh fly species obtained from DFA. Light blue represents the mean shape of the female and dark blue represents the mean shape of the male.

CV2 = 34.7%). The scatter plot from CV1 and CV2 (Fig. 5) showed large overlapping between *B. peregrina* and *B. nathani*, and between *L. pattoni* and *L. saprianovae*, whereas *L. ruficornis* and *P. dux* were observed to be clearly separated from the others. The Mahalanobis distances obtained from the pairwise comparisons of the six flesh fly species revealed highly significant differences (permutation test with 10,000 rounds in MorphoJ: $P < 0.0001$), ranging from 3.9708 (*B. nathani* and *B. peregrina*) to 13.6861 (*L. ruficornis* and *B. nathani*) (Supplementary Table S3). The cross-validation test showed a high percentage of correctly classified specimens in the majority of the comparisons (> 93.6%), except for *B. nathani* and *B. peregrina* (< 82%) (Table 5).

Table 6
Percentages of correct classification between sexes of six flesh fly species obtained from cross-validation test using permutation test with 10,000 rounds in MorphoJ.

Species	% correctly classified (no. of correctly classified/total no. of specimens)	
	Males	Females
*B. nathani***	87.0 (20/23)	100.0 (22/22)
*B. peregrina**	95.2 (20/21)	81.3 (13/16)
*L. pattoni***	100.0 (32/32)	100.0 (31/31)
*L. ruficornis***	95.9 (47/49)	100.0 (62/62)
*L. saprianovae***	100.0 (38/38)	100.0 (34/34)
*P. dux***	97.8 (45/46)	97.9 (46/47)

The significant differences between males and females based on the Mahalanobis distances, which is denoted with asterisks (permutation test with 10,000 rounds in MorphoJ: $^{**}P < 0.0001$; $^{*}P < 0.01$).

3.3. Sexual shape dimorphism (SShD)

The DFA revealed highly significant differences in the wing shape between males and females of all the six species (permutation test with 10,000 rounds in MorphoJ: $P < 0.0001$ and $P < 0.01$). Visualized shape differences between the sexes of each species showed that the females were larger than the males (Fig. 6), as seen in the displacement of landmarks 3, 4, 5, 6, 7, 8, 9, and 10. The percentage of correct classification of the males ranged from 87.0% (*B. nathani*) to 100.0% (*L. pattoni* and *L. saprianovae*), while the percentage of correct classification of the females ranged from 81.3% (*B. peregrina*) to 100% (*B. nathani*, *L. pattoni*, *L. saprianovae,* and *L. ruficornis*) (Table 6).

4. Discussion

Estimation of the allometric effects is an essential step in any study of morphological variation as regards morphometric studies because allometry can influence taxonomic studies and sexual dimorphism (Gidaszewski et al., 2009; Sontigun et al., 2017). In the present study, the allometric analysis showed wing size explains part of the variation in wing shape among species, within a species, and between sexes in most species. Although the effect of scaling relationships was very low, size correction by using residuals from the regression of shape on size was used for studying the shape variation without the size effect. Such a size correction is widely used in taxonomic studies for controlling the confounding of size in species discrimination based on shape (Sidlauskas et al., 2011; Lorenz et al., 2012; Sontigun et al., 2017).

Wing shape has proven to be a valuable tool for both species and sex discrimination in many insects (Gidaszewski et al., 2009; Jaramillo-O et al., 2015; Virginio et al., 2015; Sontigun et al., 2017). Therefore, it was not surprising that wing shape could be used to discriminate medically and forensically important flesh flies of Thailand at both the genus and the species levels. Although wing shape showed large overlapping among species in the morphospace of canonical variables, a high percentage of correct classification was found in the majority of the comparisons (81.3–100.0%). The large overlapping in the wing shape among the various species of flesh flies reveals that the wing shapes of the different species are greatly similar to each other. The same pattern was observed in the external morphology of adult flies that are extremely similar, making it very difficult to identify them: they can largely be identified based on the morphology of the external male genitalia (Kurahashi and Chaiwong, 2013; Vairo et al., 2014, 2015). Because of the high complexity of the structure of the male genitalia, it is quite difficult for non-taxonomists to perform the identification; therefore, landmark-based analysis of wings can be an alternative tool for identification of Thailand flesh fly species, as it has previously been shown in Australian Sarcophaginae (Brown and Shipp, 1978). However, for the low percentage of correct classification of some pairs of species, that is, *B. nathani* and *B. peregrina*; *P. brevicornis* and *P. scopariiformis*; *P. brevicornis* and *S. princeps*; *P. scopariiformis* and *S. antilope*; *P. scopariiformis* and *S. princeps*; *S. antilope* and *S. multivillosa*; and *S. antilope* and *S. seniorwhitei*, it is recommended that wing morphometric analysis be used in combination with identification keys or DNA analysis for accurate species identification. In comparison to molecular methods, the landmark-based analysis of wings is much easier, faster, and cheaper. Moreover, this method requires only basic training and non-damaged wings for analysis.

Female flesh flies are more abundant than males on carrion-baited traps and carcasses (Martín-Vega and Baz, 2013; Szpila et al., 2015; Villet et al., 2017); for that reason, correct identification of females is very important. However, adult females are extremely similar in appearance and identification keys for females are rare (Pape, 1996; Richet et al., 2011; Vairo et al., 2015), making the identification of females problematic and often impossible for those who are not specialists in the taxonomy of this group. This present study revealed that the wing shape of females can be used to separate flesh fly species with high reliability, particularly in *L. pattoni*, *L. ruficornis*, *L. saprianovae*, and *P. dux* which showed a very high percentage of correct classification (> 93.6%). This result indicates that wing shape is a useful structure for species identification of female flesh flies.

There were significant differences observed in the wing shape between sexes in the six flesh fly species analyzed, which indicates that phenotypic expression of wing shape is sex-specific and that wing morphometric analysis can be an alternative tool for sex discrimination. This result is consistent with the finding reported in other insects (Gidaszewski et al., 2009; Hall et al., 2014; Virginio et al., 2015). In addition, female wings were found to be larger than male wings in all the analyzed species. Similar findings have been observed in other insects, for example, mosquitoes (Virginio et al., 2015), screwworm flies (Lyra et al., 2010), and moths (de Camargo et al., 2015). Several studies have demonstrated that allometry is an important factor of sexual shape dimorphism (SShD) in the wings (Gidaszewski et al., 2009; Sontigun et al., 2017). However, some studies have also shown that allometry has no impact on SShD in the wings as these analyses of dimorphism with and without the effects of allometry yielded similar results (de Camargo et al., 2015; Virginio et al., 2015). In the present study, the results show that SShD in the wings was not the effect of allometry because the allometry was removed before the shape analyses, suggesting that factors other than allometry may have an effect on SShD in flesh flies. In this regard, the differences between the sexes in flight behavior are likely to influence the evolution of SShD in the wings of flesh flies (Paquette et al., 2008). However, knowledge of flight behavior in relation to wing morphology of flesh flies is needed in order to explain the SShD of flesh flies.

In conclusion, the results of this study demonstrate that landmark-based geometric morphometric analysis of wings is a reliable method for classifying flesh flies at both the genus and the species levels, particularly by non-taxonomists. The wing shape of females can also be used for species discrimination with high reliability; females are generally difficult to identify using traditional diagnostic characteristics due to lack of identification keys for females. Similarly, wing morphometric analysis can be an additional method used for identifying the sex of flesh flies.

Acknowledgment

This research was financially supported by the Diamond Research Grant (PAR-2560-04663) of the Faculty of Medicine, Chiang Mai University.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.actatropica.2018.12.

011.

References

Arnqvist, G., Mårtensson, T., 1998. Measurement error in geometric morphometrics: empirical strategies to assess and reduce its impact on measures of shape. Acta Zool. Academ. Sci. Hung. 44, 73–96.

Bänziger, H., Pape, T., 2004. Flowers, feces and cadavers: natural feeding and laying habitats of flesh flies in Thailand (Diptera: Sarcophagidae, *Sarcophaga* spp.). J. Nat. Hist. 38, 1677–1694.

Bookstein, F.L., 1991. Morphometric Tools for Landmark Data: Geometry and Biology. Cambridge University Press, New York.

Brown, K.R., Shipp, E., 1978. Wing morphometric analysis of Australian Sarcophaginae (Diptera: Sarcophagidae). Syst. Entomol. 3, 179–188.

Bubliy, O.A., Tcheslavskaia, K.S., Kulikov, A.M., Lazebny, O.E., Mitrofanov, V.G., 2008. Variation of wing shape in the *Drosophila virilis* species group (Diptera: Drosophilidae). J. Zool. Syst. Evol. Res. 46, 38–47.

Cárdenas, R.E., Hernández-L, N., Barragán, A.R., Dangles, O., 2013. Differences in morphometry and activity among tabanid fly assemblages in an Andean Tropical Montane Cloud Forest: indication of altitudinal migration? Biotropica 45, 63–72.

Carvajal, T.M., Hernandez, L.F., Ho, H.T., Cuenca, M.G., Orantia, B.M., Estrada, C.R., Viacrusis, K.M., Amalin, D.M., Watanabe, K., 2016. Spatial analysis of wing geometry in dengue vector mosquito, *Aedes aegypti* (L.) (Diptera: Culicidae), populations in Metropolitan Manila, Philippines. J. Vector Borne Dis. 53, 127–135.

Chaiwong, T., Srivoramas, T., Sukontason, K., Sanford, M.R., Moophayak, K., Sukontason, K.L., 2012. Survey of the synanthropic flies associated with human habitations in Ubon Ratchathani province of northeast Thailand. J. Parasitol. Res. 2012, 1–9.

Chaiwong, T., Tem-Eiam, N., Limpavithayakul, M., Boongunha, N., Poolphol, W., Sukontason, K.L., 2014. Aural myiasis caused by *Parasarcophaga* (*Liosarcophaga*) *dux* (Thomson) in Thailand. Trop. Biomed. 31, 496–498.

Chiewchanvit, S., Chaithong, U., Sanit, S., Samerjai, C., Sukontason, K., 2017. Dermal myiasis caused by the flesh fly, *Parasarcophaga* (*Liosarcophaga*) *dux* (Thomson, 1896) (Diptera: Sarcophagidae) at the site of a malignant melanoma: a case report. Southeast Asian J. Trop. Med. Public Health 48, 184–188.

Chigusa, Y., Shinonaga, S., Honda, M., Kanma, W., Kakinuma, H., Kirinoki, M., Matsuda, H., 2005. Vaginal myiasis due to *Sarcophaga peregrina* (Diptera: Sarcophagidae) on a patient with atrial fibrillation, cerebral infarction and leg amputation. Med. Entomol. Zool. 56, 247–249.

de Camargo, W.R., de Camargo, N.F., Corrêa Ddo, C., de Camargo, A.J., Diniz, I.R., 2015. Sexual dimorphism and allometric effects associated with the wing shape of seven moth species of Sphingidae (Lepidoptera: Bombycoidea). J. Insect Sci. 15, 107.

Ferraz, A.C.P., Proença, B., Gadelha, B.Q., Faria, L.M., Barbalho, M.G.M., Aguiar-Coelho, V.M., Lessa, C.S.S., 2010. First record of human myiasis caused by association of the species *Chrysomya megacephala* (Diptera: Callipliorididae), *Sarcophaga* (*Liopygia*) *ruficornis* (Diptera: Sarcophagidae), and *Musca domestica* (Diptera: Muscidae). J. Med. Entomol. 47, 487–490.

Getachew, S., Gebre-Michael, T., Erko, B., Balkew, M., Medhin, G., 2007. Non-biting cyclorrhaphan flies (Diptera) as carriers of intestinal human parasites in slum areas of Addis Ababa. Ethiopia. Acta Trop. 103, 186–194.

Giangaspero, A., Marangi, M., Balotta, A., Venturelli, C., Szpila, K., Di Palma, A., 2017. Wound myiasis caused by *Sarcophaga* (*Liopygia*) *argyrostoma* (Robineau-Desvoidy) (Diptera: Sarcophagidae): additional evidences of the morphological identification dilemma and molecular investigation. Sci. World J. 2017, 9064531.

Gidaszewski, N.A., Baylac, M., Klingenberg, C.P., 2009. Evolution of sexual dimorphism of wing shape in the *Drosophila melanogaster* subgroup. BMC Evol. Biol. 9, 110.

Graczyk, T.K., Knight, R., Tamang, L., 2005. Mechanical transmission of human protozoan parasites by insects. Clin. Microbiol. Rev. 18, 128–132.

Grzywacz, A., Ogiela, J., Tofilski, A., 2017. Identification of Muscidae (Diptera) of medico-legal importance by means of wing measurements. Parasitol. Res. 116, 1495–1504.

Guo, Y.D., Cai, J.F., Xiong, F., Wang, H.J., Wen, J.F., Li, J.B., Chen, Y.Q., 2012. The utility of mitochondrial DNA fragments for genetic identification of forensically important sarcophagid flies (Diptera: Sarcophagidae) in China. Trop. Biomed. 29, 51–60.

Hall, M.J.R., MacLeod, N., Wardhana, A.H., 2014. Use of wing morphometrics to identify populations of the Old World screwworm fly, *Chrysomya bezziana* (Diptera: Calliphoridae): a preliminary study of the utility of museum specimens. Acta Trop. S49–S55 138S.

Jaramillo-O, N., Dujardin, J.P., Calle-Londoño, D., Fonseca-González, I., 2015. Geometric morphometrics for the taxonomy of 11 species of *Anopheles* (*Nyssorhynchus*) mosquitoes. Med. Vet. Entomol. 29, 26–36.

Jordaens, K., Sonet, G., Richet, R., Dupont, E., Braet, Y., Desmyter, S., 2013. Identification of forensically important *Sarcophaga* species (Diptera: Sarcophagidae) using the mitochondrial COI gene. Int. J. Legal Med. 127, 491–504.

Kavitha, R., Nazni, W.A., Tan, T.C., Lee, H.L., Azirun, M.S., 2013. Review of forensically important entomological specimens collected from human cadavers in Malaysia (2005-2010). J. Forensic Leg. Med. 20, 480–482.

Klingenberg, C.P., 2011. MorphoJ: an integrated software package for geometric morphometrics. Mol. Ecol. Resour. 11, 353–357.

Kurahashi, H., Chaiwong, T., 2013. Keys to the flesh flies of Thailand, with description of a new species of *Robineauella* Enderlein (Diptera: Sarcophagidae). Med. Entomol. Zool. 64, 83–101.

Lorenz, C., Marques, T.C., Sallum, M.A.M., Suesdek, L., 2012. Morphometrical diagnosis of the malaria vectors *Anopheles cruzii*, *An. homunculus* and *An. bellator*. Parasit. Vectors 5, 257.

Lyra, M.L., Hatadani, L.M., de Azeredo-Espin, A.M., Klaczko, L.B., 2010. Wing morphometry as a tool for correct identification of primary and secondary New World screwworm fly. Bull. Entomol. Res. 100, 19–26.

Martín-Vega, D., Baz, A., 2013. Sex-biased captures of sarcosaprophagous Diptera in carrion-baited traps. J. Insect Sci. 13, 14.

Meiklejohn, K.A., Dowton, M., Pape, T., Wallman, J.F., 2013. A key to the Australian Sarcophagidae (Diptera) with special emphasis on *Sarcophaga* (sensu lato). Zootaxa 3680, 148–189.

Muñoz-Muñoz, F., Talavera, S., Carpenter, S., Nielsen, S.A., Werner, D., Pagès, N., 2014. Phenotypic differentiation and phylogenetic signal of wing shape in western European biting midges, *Culicoides* spp., of the subgenus *Avaritia*. Med. Vet. Entomol. 28, 319–329.

Pape, T., 1996. Catalogue of the Sarcophagidae of the world (Insecta: Diptera). Mem Entomol. Int. 8, 1–558.

Paquette, C., Joplin, K., Seier, E., Peyton, J.T., Moore, D., 2008. Sex-specific differences in spatial behaviour in the flesh fly *Sarcophaga crassipalpis*. Physiol. Entomol. 33, 382–388.

Pepinelli, M., Spironello, M., Currie, D.C., 2013. Geometric morphometrics as a tool for interpreting evolutionary transitions in the black fly wing (Diptera: Simuliidae). Zool. J. Linn. Soc. 169, 377–388.

Perre, P., Jorge, L.R., Lewinsohn, T.M., Zucchi, R.A., 2014. Morphometric differentiation of fruit fly pest species of the *Anastrepha fraterculus* Group (Diptera: Tephritidae). Ann. Entomol. Soc. Am. 107, 490–495.

Richet, R., Blackith, R.M., Pape, T., 2011. *Sarcophaga* of France (Diptera: Sarcophagidae). Pensoft Series Faunistica, Sofia, Bulgaria.

Sidlauskas, B.L., Mol, J.H., Vari, R.P., 2011. Dealing with allometry in linear and geometric morphometrics: a taxonomic case study in the *Leporinus cylindriformis* group (Characiformes: Anostomidae) with description of a new species from Suriname. Zool. J. Linn. Soc. 162, 103–130.

Sontigun, N., Sukontason, K.L., Zajac, B.K., Zehner, R., Sukontason, K., Wannasan, A., Amendt, J., 2017. Wing morphometrics as a tool in species identification of forensically important blow flies of Thailand. Parasit. Vectors 10, 229.

Sukchit, M., Deowanish, S., Butcher, B.A., 2015. Decomposition stages and carrion insect succession on dressed hanging pig carcasses in Nan province, northern Thailand. Trop. Nat. Hist. 15, 137–153.

Sukontason, K., Narongchai, P., Kanchai, C., Vichairat, K., Sribanditmongkol, P., Bhoopat, T., Kurahashi, H., Chockjamsai, M., Piangjai, S., Bunchu, N., Vongvivach, S., Samai, W., Chaiwong, T., Methanitikorn, R., Ngern-Klun, R., Sripakdee, D., Boonsriwong, W., Siriwattanarungsee, S., Srimuangwong, C., Hanterdsith, B., Chaiwan, K., Srisuwan, C., Upakut, S., Moopayak, K., Vogtsberger, R.C., Olson, J.K., Sukontason, K.L., 2007. Forensic entomology cases in Thailand: a review of cases from 2000 to 2006. Parasitol. Res. 101, 1417–1423.

Sukontason, K., Bunchu, N., Chaiwong, T., Moophayak, K., Sukontason, K.L., 2010. Forensically important flesh fly species in Thailand: morphology and developmental rate. Parasitol. Res. 106, 1055–1064.

Szpila, K., Mądra, A., Jarmusz, M., Matuszewski, S., 2015. Flesh flies (Diptera: Sarcophagidae) colonising large carcasses in Central Europe. Parasitol. Res. 114, 2341–2348.

Tan, S.H., Rizman-Idid, M., Mohd-Aris, E., Kurahashi, H., Mohamed, Z., 2010. DNA-based characterisation and classification of forensically important flesh flies (Diptera: Sarcophagidae) in Malaysia. Forensic Sci. Int. 199, 43–49.

Vairo, K.P., Ururahy-Rodrigues, A., Moura, M.O., de Mello-Patiu, C.A., 2014. Sarcophagidae (Diptera) with forensic potential in Amazonas: a pictorial key. Trop. Zool. 27, 140–152.

Vairo, K.P., Moura, M.O., de Mello-Patiu, C.A., 2015. Comparative morphology and identification key for females of nine Sarcophagidae species (Diptera) with forensic importance in Southern Brazil. Rev. Bras. Entomol. 59, 177–187.

Villet, M.H., Clitheroe, C., Williams, K.A., 2017. The temporal occurrence of flesh flies (Diptera, Sarcophagidae) at carrion-baited traps in Grahamstown, South Africa. Afr. Invertebr. 58, 1–8.

Virginio, F., Vidal, P.O., Suesdek, L., 2015. Wing sexual dimorphism of pathogen-vector culicids. Parasit. Vectors 8, 159.

Vitta, A., Pumidonming, W., Tangchaisuriya, U., Poodendean, C., Nateeworanart, S., 2007. A preliminary study on insects associated with pig (*Sus scrofa*) carcasses in Phitsanulok, northern Thailand. Trop. Biomed. 24, 1–5.

Wilke, A.B., Christe Rde, O., Multini, L.C., Vidal, P.O., Wilk-da-Silva, R., de Carvalho, G.C., Marrelli, M.T., 2016. Morphometric wing characters as a tool for mosquito identification. PLoS One 11 e0161643.